

kmhelpers: A Python toolkit for automated management of genomic indexes

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Summary

Large-scale genomic sequence search has become a central problem in modern bioinformatics (Karasikov et al., 2025; Marchet, 2024). A widely used family of methods relies on Bloom filters (BF) (Bloom, 1970) to record, for each indexed sample, which k -mers (words of length k) it contains; a query then reports the set of samples containing each queried k -mer. Building such an index over many samples is not a single operation but a chain of interdependent steps, and each step demands specialist knowledge: sizing each Bloom filter to its sample, configuring the third-party components used internally by kmindex (such as the findere (Robidou & Peterlongo, 2021) approximate-membership layer), distributing data and computation, bounding peak RAM, and grouping samples into sub-indexes so that the final index stays small without slowing queries. An error at any step can invalidate downstream results, waste significant computation, or produce an oversized index.

We present kmhelpers, an open-source Python toolkit that automates this entire workflow for indexes built with kmindex (Lemane et al., 2024). It hides the technical decisions listed above behind a command-line interface (CLI) and a matching Python API that cover every stage of the k -mer index lifecycle, from raw samples up to queries, including the federation of independently built indexes into a single queryable resource.

Statement of Need

k -mer-based sequence databases built with tools such as kmindex (Lemane et al., 2024) and kmtricks (Lemane et al., 2022) answer queries on large-scale samples in genomics, metagenomics, and population studies. They were recently used to index 50 petabytes of raw sequencing data (Chikhi et al., 2024). Yet a wide gap separates having the indexing software installed from running a reproducible, end-to-end, size-optimised indexing workflow.

Crossing that gap currently requires a user to master, simultaneously: Bloom filter sizing as a function of each sample's k -mer cardinality; the configuration of internal components of kmindex; the distribution of data and build jobs across the available compute; the control of peak memory; and the partitioning of samples into sub-indexes that balances index size against query speed. Only specialists hold all of this knowledge at once.

kmhelpers removes that barrier. It makes the complete process of building, updating, and querying a kmindex search engine accessible to any group that holds a large, evolving genomic dataset (research teams, sequencing platforms, hospitals, or data-production centres) whether the resulting index is kept private or shared. This opens to non-specialists both the creation and maintenance of private indexes and the contribution of sub-indexes to larger, federated search engines.

Formally, the input is a set $\mathcal{S} = \{S_1, \dots, S_n\}$ of n genomic samples of various sizes; each S_i is either a raw sequencing dataset or assembled sequences. The output is a kmindex index

subject to two user-defined limits: (1) the maximum allowed false-positive rate, and (2) the maximum number of sub-indexes, each a file storing a set of BFs as a matrix. `kmhelpers` orchestrates every step between input and output, automating the parametrisation and the data distribution.

State of the Field

Several tools tackle large-scale sequence search with k -mer-based data structures. BIGSI (Bradley et al., 2019) and COBS (Bingmann et al., 2019) use compressed Bloom filter matrices to answer presence/absence queries across collections of sequencing experiments. HowDeSBT (Harris & Medvedev, 2020) and Mantis (Pandey et al., 2018) rely on sequence Bloom trees and counting quotient filters, respectively. More recent colored- k -mer indexes such as MetaGraph (Karasikov et al., 2025) and Fulgor (Fan et al., 2024) push scalability further; see (Marchet, 2024) for a survey. `kmindex`, which `kmhelpers` wraps, builds on the `kmtricks` (Lemane et al., 2022) counting pipeline and is designed for efficient querying of large sequence collections.

`kmhelpers` does not compete with any of these approaches at the algorithmic level. Its contribution is orthogonal and, importantly, is not merely a pipeline script. The difficulty here lies not in chaining steps but in the individual components and how they are orchestrated. `kmhelpers` implements the decisions a `kmindex` expert would otherwise make by hand: profile computes Bloom filter sizes and sample-to-sub-index assignments that minimise total index size under a false-positive-rate constraint; plan estimates disk and memory requirements before any build starts; and registry federates independently built sub-indexes into one logical index. No dedicated automation layer of this kind existed for `kmindex` prior to `kmhelpers`.

Software design

Background: the sub-index structure

A Bloom-filter index stores all BFs of the same size as a single (row-major) matrix in one file. In such a matrix each column is one BF (one sample) and each row records the presence (1) or absence (0) of a k -mer across all samples sharing that filter size, assuming a common hash function. A complete index is therefore a set of such matrices, each one a *sub-index* holding all BFs of a given size.

The difficulty is that each BF size must match the number of items it stores, here the distinct k -mers of a sample. If all samples had the same number of distinct k -mers, every BF would share one size and a single matrix would suffice. This is the ideal case, where only one file is opened per query and one matrix row gives the answer across all n samples. In practice, sample sizes differ by several orders of magnitude. Sizing every BF for the largest sample wastes enormous storage. Giving each sample its own single-column matrix minimises storage but forces a query to open n files (potentially millions), severely slowing down queries. `kmhelpers` takes the middle ground: the user fixes the maximum number of sub-indexes, and the profile step chooses the per-sub-index BF size that minimises total index size under that constraint, before to distribute each input sample to its respective sub-index.

The pipeline

`kmhelpers` exposes the index lifecycle as a sequence of commands, illustrated Figure 1:

- **list** — recursively discovers all samples in a given directory and counts each sample's distinct k -mers using `ntCard` (Mohamadi et al., 2017) (unless the counts are provided by the user).
- **profile** — determines the best set of sub-index BF sizes given the user-defined maximum number of sub-indexes and target false-positive rate.

- **compose** — assigns each sample to its sub-index and generates the *files-of-files* describing the data origin of each sub-index.
- **plan** — validates sample files, available disk space, and memory upfront, and emits ready-to-execute pipeline scripts.
- **apply** — builds all sub-indexes by invoking `kmindex`, with span-level and name-level filtering.
- **compress** — optional ZSTD-based block compression of the index (?).
- **registry** — registers several sub-indexes (built locally or anywhere accessible) into one logical index, redirecting each query to the relevant sub-indexes at query time.

Once an index is built, `kmhelpers` also answers queries (query). Multi-step workflows can be described as declarative YAML pipelines (pipeline) and executed in a single command.

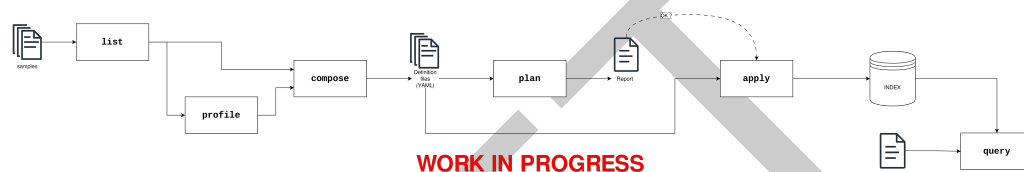


Figure 1: Overview of the `kmhelpers` workflow

Implementation

`kmhelpers` is implemented in Python (≥ 3.8) and distributed via Conda, which installs its bioinformatics dependencies (`kmindex`, `ntCard`) automatically. The CLI is built with Click (Ronacher & Click Contributors, 2023), and the package exposes a public Python API covering all CLI functionality. Together, the declarative YAML index-definition format and the plan/apply separation make *k*-mer indexing workflows accessible to researchers who are not experts in the underlying data structures, while remaining flexible enough for large-scale production use.

Research impact statement

Applying `kmhelpers` to the “Tree of Life” dataset

As part of the broader “Earth Biogenome Project” (Gupta, 2022), the “Darwin Tree of Life Project” (Life Project Consortium, 2022) aims to deliver at-scale genome sequences of unprecedented quality for Britain and Ireland. It collects robustly identified specimens, performs sequencing, and generates high-quality, curated public assemblies.

This project is a perfect use case for applying `kmhelpers`: from input data – here a mixture of raw sequencing reads and assembled genomes – to a final index powering a CLI search engine (currently private; XXX).

Concretely, `kmhelpers` was applied to 7394 samples. The index is continuously updated as new sequencing data are progressively added.

Using `kmhelpers` for updating the logan-search search engine

The logan-search search engine (?) indexes approximately 50 PB of sequence data from SRA (?). Since its creation, the SRA database size doubled. The logan-search index will be updated using `kmhelper`, also optimizing the sub-indexes designs.

Availability and documentation

kmhelpers is released under the GNU General Public License v3.0 and is available at <https://github.com/seblins/kmhelpers>. The full documentation is available at <https://seblins.github.io/kmhelpers/>.

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AI Usage Disclosure

AI assistance was used for constrained tasks (drafting, editing, code suggestions) under strict human review at every stage. AI provider used: Claude (Anthropic, 2025).

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